General (k,h) in two dimensions: $Z(n) = \dfrac{n^{-p_1/2}\, b^{k_2(p_2-p_1)}}{k_1 k_2} \int_0^{\sqrt{n}\, b^{k_1+k_2}} x^{p_1-p_2-1} F_{p_2-1}(x)\, dx$

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Abstract

Units 35–43 kept $k = (1,\ldots,1)$. This unit proves the exact two-dimensional reduction for arbitrary $k = (k_1,k_2)$ and $h = (h_1,h_2)$, in the form proposed by the GPT-6 Astra consult (identity (5) of `tide-log/gpt6_gen2d_v1.md`): with $p_i = (h_i{+}1)/k_i$ and the weighted primitives $F_\gamma(x) = \int_0^x t^\gamma f(t)\,dt$ of the quadratic kernel,

$$Z(n) = \frac{1}{k_1 k_2}\, n^{-p_1/2}\, b^{k_2(p_2-p_1)} \int_0^{\sqrt{n}\, b^{k_1+k_2}} x^{p_1-p_2-1} F_{p_2-1}(x)\, dx$$

for every $n > 0$. The candidate exponents of `thm:TaylorTree` are $p_i/2$; the integral converges iff $p_1 < p_2$, grows logarithmically iff $p_1 = p_2$, and is dominated by the other coordinate iff $p_1 > p_2$. Verified numerically to machine precision in four configurations including $k_i > 1$. Zero `sorry`/`axiom`.

1 Setting

Two changes of variables suffice. The chart power substitution $\int_0^b u^h g(u^k)\,du = \frac{1}{k}\int_0^{b^k} s^{(h+1)/k-1} g(s)\,ds$ converts each coordinate's monomial weight into a real power of the new variable, and the power-weighted scaling $\int_0^L s^\gamma G(ds)\,ds = d^{-(\gamma+1)}\int_0^{dL} x^\gamma G(x)\,dx$ moves the n- and u-dependence out of the kernel. Applied inside-out, the inner integral becomes $(1/k_2)\, u^{h_1} (\sqrt{n}\, u^{k_1})^{-p_2} F_{p_2-1}(\sqrt{n}\, u^{k_1} b^{k_2})$, and the outer integral is one more substitution and one more scaling.

2 The things formalised

```
theorem integral_Ioc_pow_mul_comp_pow (h k :  ) (b :  ) (g :   →  )
    (hk : 0 < k) (hb : 0 < b) :
    (  u in Ioc 0 b, u ^ h * g (u ^ k))
      = 1 / (k :  ) *   s in Ioc 0 (b ^ k), s ^ (((h :  ) + 1) / k - 1) * g s
theorem integral_Ioc_rpow_mul_comp_mul_left (  d L :  ) (G :   →  )
    (hd : 0 < d) (hL : 0   L) :
    (  s in Ioc 0 L, s ^   * G (d * s))
      = d ^ (-(  + 1)) *   x in Ioc 0 (d * L), x ^   * G x
noncomputable def weightedPrimitive (  a   x :  ) :   :=
   t in Ioc 0 x, t ^   * quadKernel   a t
```

```
theorem twoDGeneral_eq ... (hn : 0 < n) (hb : 0 < b) (hk  : 0 < k ) (hk  : 0 < k ) :
    twoDGeneral   a b n h  h  k  k
      = 1 / ((k  :  ) * k )
        * (n ^ (-(((h  :  ) + 1) / k  / 2))
          * b ^ ((k  :  ) * (((h  :  ) + 1) / k  - ((h  :  ) + 1) / k )))
        *   x in Ioc 0 (Real.sqrt n * b ^ (k  + k )),
            x ^ (((h  :  ) + 1) / k  - ((h  :  ) + 1) / k  - 1)
              * weightedPrimitive   a (((h  :  ) + 1) / k  - 1) x
```

3 Proof strategy

The power substitution is Mathlib's half-line lemma `integral_comp_rpow_Ioi_of_pos` applied to the indicator of $(0, b^k]$: the pointwise identity $kx^{k-1} \cdot \frac{1}{k}(x^k)^{(h+1)/k-1} = x^h$ and the equivalence $x^k \le b^k \iff x \le b$ turn the substituted indicator into the indicator of $(0, b]$, and `setIntegral_indicator` converts both sides back to set integrals. No integrability hypotheses are needed because the lemma is a measure-preservation statement. The scaling is `intervalIntegral.integral_comp_mul_left` after the pointwise factorisation $s^\gamma = d^{-\gamma}(ds)^\gamma$. The two-dimensional theorem composes these with the kernel identity $e^{-\beta n t^2 + \beta\sqrt{n}\,ta} = f(\sqrt{n}\,t)$; the exponents p_i and the scale $d = \sqrt{n}\,b^{k_2}$ are introduced as opaque variables (via `obtain  p, hp  :  p, p = …`) so that `ring` treats them as atoms, and the constant $(\sqrt{n})^{-p_2} d^{-(p_1-p_2)} = n^{-p_1/2} b^{k_2(p_2-p_1)}$ is assembled from three one-line `rpow` identities.

4 Roadblocks and resolutions

- `integral_comp_rpow_Ioi_of_pos` takes the function implicitly on this pin; pass it as `(g := G)`. This was the only compile error.
- Folding a compound exponent with `set` would also fold it inside later-rewritten terms inconsistently; opaque `obtain` variables plus `rw [← hp]` at the two places new occurrences appear kept every atom syntactically identical for `ring`.

5 What was Mathlib, what was new

Mathlib: `integral_comp_rpow_Ioi_of_pos`, `setIntegral_indicator`, `pow_le_pow_iff_left` , `Real.mul_rpow`, `Real.rpow_natCast`, `intervalIntegral.integral_comp_mul_left`. New: the weighted primitives, both substitution lemmas in chart form, and the exact general two-dimensional reduction.

6 Lessons

A consult plan that names an explicit identity (here identity (5)) is worth checking numerically *before* formalising, and then formalising verbatim: the four numerical configurations took a minute and the Lean proof compiled on the second attempt. Substitutions on bounded charts are best done via half-line lemmas and indicators rather than interval-integral change-of-variables lemmas, whose derivative hypotheses fail at $u = 0$ for $k > 1$.

7 Outcome

The two-dimensional standard integral is now reduced exactly for every (k, h). What remains for the two-dimensional theory is the asymptotics of $\int_0^L x^{p_1-p_2-1} F_{p_2-1}$ in the three regimes; the $p_1 = p_2$ case needs $A_\gamma - F_\gamma(x) \leq A_{\gamma+1}/x$, the analogue of unit 35's $A - F(x) \leq M/x$.